Module Title:	**Infrastructure as Code (IaC)**

Module Code:	
Credits:	10
Credit Level:	9
Prerequisite Modules:	None

Hours per Week	
Lectures	**2**
Tutorials	
Lab/Studio/Practicals	**2**
Independent Learning	**13**
Total	**17**

Description:

This subject will focus on the learner's ability to plan, design and implement a programmatic solution to the creation, management, monitoring and disposal of infrastructure, systems and services. Desired state systems to ensure consistency of environments plays a strong role in this module.

Module Learning Outcomes:

On successful completion of this module the learner will be able to:

1. Automate the creation of basic infrastructure in a networked environment.
2. Create robust IaC scripts through refactoring, analysis and debugging for consistency and performance when provisioning services and infrastructure.
3. Design and apply policies for restricted access to resources during the creation of infrastructure.
4. Design, implement, and test code business objective and organizational target through pre-established SLAs and SLOs.
5. Apply best practice to the version control, security and quality assurance of IaC scripts through testing and refinement on a continuous basis.
6. Analyse a problem space, critically appraise alternative solutions, and design a resilient solution to a problem following appropriate industry defined best practice and standards.
7. Plan, design, encode, assess, refactor, and present a solution to a given business problem for the implementation of IaC.

Indicative Content:

1. Infrastructure Automation

- Scripting, Internal and External DSLs
- Multi-configuration Projects
- YAML, JSON, XML

2. Automation

- Programming Paradigms: Functional, Object Orientation, Desired State
- Parameters; complex data structures: lists, dictionaries, tuples
- Conditional statements
- Scope
- Internal and external resources
- RESTful APIs
- Modules and plugins
- Policies: security, resource, deletion
- Regular expressions for data filtering
- Templates, mapping resources

3. IaC Management

- Configuration drift, resolving drift automatically
- Rollback
- Benchmarking
- Best Practice, standards, industry requirements
- Tools and process for version control of IaC
- Logging, reporting

4. Trends

- Technology trends for IaC
- Architecture trends for IaC
- Emerging Concepts
- Alternative programming languages

Module Assessment:

Coursework	100 %
End of Semester Final Exam	%

Learning Outcome	Addressed by	
	Coursework	End of Semester Final Exam
1	√	
2	√	
3	√	
4	√	
5	√	
6	√	
7	√	

Indicative Assessment

Element No	Weighting	Type	Description	Learning Outcome Assessed
1	50%	Secure, Reliable, IaC Networked Solution	Each student will produce a solution to a given scenario requiring the creation of infrastructure resources, appropriately networked, highly available and secure. This solution should include multiple automated resources in an appropriate architect. The learner should draw conclusions on the work carried out.	1, 2,3,5,6,7
2	50%	Practical Examination	The student will analyse a broken IaC script, review suggested solutions, present an improved solution using a variety of practical techniques, applying best practice and standards as appropriate.	1,3,4,6,7

Resources:

Note: Learning resources may also be available on Blackboard.

Recommended Reading				
Author	**Year**	**Title**	**Publisher**	**ISBN**
Abuelenain, Khaled, Doyle, Jeff, Karneliuk, Anton and Jain, Vinit	2021	Network Programmability and Automation Fundamentals	Cisco Press	978-1587145148
Mishra, Ravi	2021	HashiCorp Infrastructure Automation Certification Guide: Pass the Terraform Associate exam and manage IaC to scale across AWS, Azure, and Google Cloud	Packt Publishing	978-1800565975
Lee, Thomas	2020	PowerShell 7 for IT Professionals	Wiley	978-1119644729

Other Resources

This module will rely in industry reports for best practice, trends and detailed configuration information. Extensive review of blogs, websites, online journals and conference material is expected as books often lag behind online resources regarding current trends.

Cisco DevNet

Fortinet Security Academy

F5 Digital Education

VMWare Customer Connect Learning

Webography:

ACM Digital Library: https://dl.acm.org/

IEEE Explore Digital Library: https://ieeexplore.ieee.org/Xplore/home.jsp

National Standards Authority Ireland: https://www.nsai.ie/standards/

European Union Laws & Regulations: https://eur-lex.europa.eu/homepage.html

OASIS Open: https://www.oasis-open.org/